

## NAHA AP, Japan

WMO: 479300

Lat: 26.196N

Lon: 127.646E

Elev: 4

StdP: 101.28

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.2       | 13.1 | 3.0                           | 4.7 | 13.8 | 4.0 | 5.0 | 14.3 | 14.4                  | 14.7 | 13.3 | 15.1 | 7.8                   | 350  | 0.467 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 3.9                    | 32.2            | 26.5 | 32.0 | 26.5 | 31.2 | 26.5 | 27.8                | 30.3 | 27.5 | 30.1 | 27.1 | 29.9 | 5.7                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.1                            | 22.8 | 30.0 | 26.8 | 22.4 | 29.8 | 26.2 | 21.6 | 29.4 | 88.6          | 30.8 | 87.2 | 30.6 | 86.0 | 30.7 | 29.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 13.3              | 11.6 | 10.3 | 10.8                       | 33.3 | 1.4                | 0.9 | 9.8                                                | 34.0 | 9.0        | 34.5 | 8.3        | 35.0 | 7.3        | 35.7 |
|                   |      |      | 6.8                        | 28.4 | 0.9                | 0.5 | 6.2                                                | 28.7 | 5.6        | 29.0 | 5.1        | 29.3 | 4.5        | 29.6 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 23.7   | 17.7  | 18.0  | 19.4  | 21.9  | 24.5  | 27.4  | 29.4  | 29.3  | 28.4  | 26.1  | 23.1  | 19.5 |
|                                                                                  | DBStd       | 4.65   | 2.33  | 2.61  | 2.50  | 2.18  | 1.81  | 1.73  | 1.02  | 1.09  | 1.28  | 1.70  | 2.02  | 2.35 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 112    | 41    | 35    | 19    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 14   |
|                                                                                  | CDD10.0     | 5016   | 238   | 224   | 291   | 356   | 450   | 521   | 601   | 599   | 551   | 499   | 392   | 295  |
|                                                                                  | CDD18.3     | 2086   | 21    | 26    | 52    | 108   | 192   | 271   | 343   | 341   | 301   | 241   | 142   | 51   |
|                                                                                  | CDH23.3     | 19178  | 8     | 17    | 54    | 291   | 1142  | 2824  | 4405  | 4347  | 3485  | 1989  | 558   | 58   |
|                                                                                  | CDH26.7     | 6452   | 0     | 0     | 0     | 11    | 128   | 829   | 1947  | 1894  | 1224  | 382   | 37    | 1    |
| Wind                                                                             | WSAvg       | 5.2    | 5.6   | 5.5   | 5.4   | 4.9   | 4.5   | 5.0   | 5.0   | 4.8   | 5.1   | 5.4   | 5.2   | 5.5  |
| Precipitation                                                                    | PrecAvg     | 2067   | 113   | 111   | 151   | 164   | 245   | 262   | 176   | 252   | 213   | 164   | 119   | 112  |
|                                                                                  | PrecMax     | 3326   | 275   | 336   | 420   | 525   | 637   | 861   | 510   | 674   | 1096  | 689   | 367   | 260  |
|                                                                                  | PrecMin     | 964    | 22    | 20    | 28    | 34    | 13    | 20    | 5     | 22    | 23    | 7     | 4     | 1    |
|                                                                                  | PrecStd     | 494    | 55    | 72    | 88    | 104   | 146   | 157   | 147   | 148   | 179   | 129   | 77    | 65   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 24.2  | 25.1  | 26.0  | 27.9  | 29.9  | 32.0  | 33.2  | 33.2  | 32.8  | 31.1  | 29.0  | 26.0 |
|                                                                                  |             | MCWB   | 20.0  | 21.8  | 22.4  | 24.6  | 26.2  | 27.1  | 26.7  | 26.7  | 26.3  | 25.8  | 24.5  | 21.9 |
|                                                                                  | 2%          | DB     | 23.1  | 23.9  | 24.8  | 26.9  | 28.9  | 31.1  | 32.2  | 32.2  | 31.9  | 30.1  | 27.8  | 24.9 |
|                                                                                  |             | MCWB   | 19.2  | 20.4  | 21.2  | 23.8  | 25.5  | 27.0  | 26.6  | 26.6  | 26.2  | 25.3  | 23.3  | 20.6 |
|                                                                                  | 5%          | DB     | 22.1  | 22.9  | 23.9  | 25.9  | 28.0  | 30.8  | 32.0  | 32.0  | 31.1  | 29.2  | 26.9  | 23.9 |
|                                                                                  |             | MCWB   | 18.1  | 19.6  | 20.5  | 23.0  | 24.7  | 27.0  | 26.6  | 26.6  | 26.1  | 24.5  | 22.6  | 19.6 |
|                                                                                  | 10%         | DB     | 21.1  | 22.0  | 23.0  | 25.0  | 27.1  | 30.0  | 31.2  | 31.2  | 30.8  | 28.8  | 26.0  | 23.0 |
|                                                                                  |             | MCWB   | 17.3  | 18.9  | 19.9  | 22.2  | 24.1  | 26.7  | 26.6  | 26.6  | 26.0  | 24.2  | 22.1  | 19.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 21.5  | 22.7  | 23.0  | 25.1  | 26.7  | 28.1  | 28.2  | 28.2  | 27.7  | 26.7  | 25.4  | 22.9 |
|                                                                                  |             | MCDB   | 23.4  | 24.1  | 24.8  | 26.8  | 28.4  | 30.2  | 30.6  | 30.7  | 30.4  | 29.5  | 27.8  | 24.9 |
|                                                                                  | 2%          | WB     | 20.1  | 21.5  | 22.3  | 24.4  | 26.1  | 27.6  | 27.7  | 27.7  | 27.1  | 26.1  | 24.3  | 21.7 |
|                                                                                  |             | MCDB   | 22.2  | 23.1  | 24.0  | 26.0  | 27.8  | 29.7  | 30.3  | 30.4  | 30.0  | 28.9  | 26.5  | 23.7 |
|                                                                                  | 5%          | WB     | 18.8  | 20.4  | 21.4  | 23.6  | 25.6  | 27.2  | 27.4  | 27.3  | 26.7  | 25.6  | 23.6  | 20.6 |
|                                                                                  |             | MCDB   | 21.4  | 22.2  | 23.2  | 25.3  | 27.3  | 29.3  | 30.2  | 30.2  | 29.7  | 28.4  | 26.0  | 23.0 |
|                                                                                  | 10%         | WB     | 17.8  | 19.2  | 20.5  | 22.8  | 25.0  | 26.8  | 27.1  | 27.0  | 26.5  | 25.1  | 22.7  | 19.5 |
|                                                                                  |             | MCDB   | 20.7  | 21.4  | 22.4  | 24.5  | 26.7  | 28.9  | 30.0  | 30.1  | 29.4  | 27.8  | 25.5  | 22.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.2   | 4.3   | 4.4   | 4.3   | 4.0   | 3.7   | 3.9   | 4.0   | 4.1   | 4.1   | 4.0   | 4.1  |
|                                                                                  |             | MCDBR  | 5.1   | 5.1   | 4.9   | 4.7   | 4.3   | 3.9   | 4.3   | 4.4   | 4.5   | 4.5   | 4.5   | 4.8  |
|                                                                                  |             | MCWBR  | 3.9   | 3.8   | 3.8   | 3.3   | 2.7   | 1.9   | 2.0   | 2.0   | 2.2   | 2.5   | 2.9   | 3.5  |
|                                                                                  | 5% WB       | MCDBR  | 4.6   | 4.6   | 4.5   | 4.3   | 4.0   | 3.6   | 3.8   | 4.0   | 4.1   | 4.1   | 4.2   | 4.3  |
|                                                                                  |             | MCWBR  | 4.3   | 4.4   | 4.5   | 3.6   | 3.0   | 2.2   | 2.1   | 2.1   | 2.2   | 2.5   | 3.3   | 3.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.462  | 0.491 | 0.565 | 0.573 | 0.516 | 0.473 | 0.474 | 0.492 | 0.487 | 0.486 | 0.469 | 0.461 |      |
|                                                                                  | taud        | 2.126  | 2.064 | 1.892 | 1.932 | 2.157 | 2.340 | 2.332 | 2.270 | 2.253 | 2.196 | 2.201 | 2.134 |      |
|                                                                                  | Ebn at Noon | 772    | 782   | 746   | 753   | 795   | 824   | 823   | 808   | 803   | 779   | 762   | 752   |      |
|                                                                                  | Edh at Noon | 138    | 157   | 196   | 193   | 154   | 127   | 128   | 136   | 135   | 136   | 127   | 132   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.43   | 3.08  | 3.79  | 4.43  | 4.95  | 5.48  | 6.50  | 5.96  | 5.22  | 4.12  | 3.01  | 2.46  |      |
|                                                                                  | RadStd      | 0.29   | 0.46  | 0.36  | 0.45  | 0.64  | 0.50  | 0.53  | 0.41  | 0.40  | 0.36  | 0.28  | 0.24  |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only           | N/A   | N/A     | N/A    | N/A     | +0.36 | +0.42 | N/A         | N/A     | N/A     | N/A     |
| Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | +0.41 | +0.53 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page